Technical Note: Design Theory for the Butterworth-Optimized Photodiode Transimpedance Amplifier

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Abstract

This document provides a self-contained description of the mathematical theory implemented in the photodiode transimpedance amplifier (TIA) design tool. It covers the single-pole opamp model, the derivation of the closed-loop second-order transfer function, the exact Butterworth-optimum feedback capacitance without the common $\omega_c R_f C_f \gg 1$ approximation, the $-3\,\mathrm{dB}$ bandwidth, the noise equivalent bandwidth for a second-order Butterworth response, the piecewise noise-gain integration for opamp voltage noise, and the shot/thermal noise analysis. All algebraic steps are shown explicitly.

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1 Circuit model and assumptions

1.1 Circuit topology

The TIA is an inverting opamp configuration with a photodiode at the input. The non-inverting input is connected to ground.

- I_{pd}: photocurrent from the photodiode (signal source, modelled as ideal current source)
- R_{f}, C_{f}: feedback resistance and capacitance (in parallel), connected from the output to the inverting input
- C_{i}: total input capacitance at the inverting node, $C_{\mathrm{i}} = C_{\mathrm{d}} + C_{\mathrm{icm}} + C_{\mathrm{id}}$
  - C_{d}: photodiode junction capacitance
  - C_{icm}: opamp common-mode input capacitance
  - C_{id}: opamp differential input capacitance
- C_{s}: stray (parasitic) capacitance at the inverting node to ground

1.2 Opamp model

The opamp is modelled with a single-pole open-loop gain:

$$A(s) = \frac{A_0\,\omega_{\mathrm{p}}}{s + \omega_{\mathrm{p}}} \tag{1}$$

where A_0 is the DC open-loop gain and ω_{p} is the dominant pole frequency. For frequencies well above the dominant pole ($s \gg \omega_{\mathrm{p}}$, i.e. $f \gg f_{\mathrm{p}}$, typically satisfied for $f > 10\,\mathrm{Hz}$), this simplifies to:

$$A(s) \approx \frac{A_0\,\omega_{\mathrm{p}}}{s} = \frac{\omega_{\mathrm{c}}}{s} \tag{2}$$

where $\omega_{\mathrm{c}} = A_0\,\omega_{\mathrm{p}} = 2\pi f_{\mathrm{c}}$ is the gain-bandwidth product (GBW) in $\mathrm{rad\,s^{-1}}$. This "integrator model" is the standard approximation used throughout TIA design literature and is accurate across the entire frequency range of interest (kHz–MHz).

2 Closed-loop transimpedance transfer function

Goal. Derive the closed-loop transimpedance:

$$Z_{\mathrm{TI}}(s) \equiv \frac{V_{\mathrm{out}}(s)}{I_{\mathrm{pd}}(s)}.$$

Step 1: Define the feedback impedance. R_{f} and C_{f} are in parallel:

$$Z_{\mathrm{f}} = R_{\mathrm{f}} \parallel \frac{1}{sC_{\mathrm{f}}} = \frac{R_{\mathrm{f}}}{1 + sR_{\mathrm{f}}C_{\mathrm{f}}} \tag{3}$$

Step 2: Opamp constraint. With the non-inverting input grounded:

$$V_{\mathrm{out}} = -A(s) \cdot V^- = -\frac{\omega_{\mathrm{c}}}{s}\,V^-$$

Solving for V^-:

$$V^- = -\frac{s}{\omega_{\mathrm{c}}}\,V_{\mathrm{out}} \tag{4}$$

Step 3: Kirchhoff's current law at the inverting node. All currents leaving the inverting node must sum to zero. The current into the node from the photodiode is I_{pd}. The currents leaving the node are:

- Through the feedback network to the output: $(V^- - V_{\mathrm{out}})/Z_{\mathrm{f}}$
- Through the total shunt capacitance $(C_{\mathrm{i}} + C_{\mathrm{s}})$ to ground: $V^- \cdot s(C_{\mathrm{i}} + C_{\mathrm{s}})$

KCL:

$$I_{\mathrm{pd}} + \frac{V^- - V_{\mathrm{out}}}{Z_{\mathrm{f}}} + V^- \cdot s(C_{\mathrm{i}} + C_{\mathrm{s}}) = 0$$

Note the sign convention: current flowing from the node through Z_{f} toward V_{out} is $(V^- - V_{\mathrm{out}})/Z_{\mathrm{f}}$. Rearranging with current into the node on the left:

$$I_{\mathrm{pd}} = \frac{V_{\mathrm{out}} - V^-}{Z_{\mathrm{f}}} - V^- \cdot s(C_{\mathrm{i}} + C_{\mathrm{s}}) \tag{5}$$

Step 4: Substitute the opamp constraint. Substitute (3) and (4) into (5).

First, compute $V_{\mathrm{out}} - V^-$:

$$V_{\mathrm{out}} - V^- = V_{\mathrm{out}}\left(1 + \frac{s}{\omega_{\mathrm{c}}}\right) = V_{\mathrm{out}} \cdot \frac{\omega_{\mathrm{c}} + s}{\omega_{\mathrm{c}}} \tag{6}$$

Substitute into the feedback current term:

$$\frac{V_{\mathrm{out}} - V^-}{Z_{\mathrm{f}}} = V_{\mathrm{out}} \cdot \frac{\omega_{\mathrm{c}} + s}{\omega_{\mathrm{c}}} \cdot \frac{1 + sR_{\mathrm{f}}C_{\mathrm{f}}}{R_{\mathrm{f}}} \tag{7}$$

The shunt capacitance current term:

$$-V^- \cdot s(C_{\mathrm{i}} + C_{\mathrm{s}}) = \frac{s^2(C_{\mathrm{i}} + C_{\mathrm{s}})}{\omega_{\mathrm{c}}} V_{\mathrm{out}} \tag{8}$$

Step 5: Combine. Substituting (7) and (8) into (5):

$$I_{\mathrm{pd}} = V_{\mathrm{out}} \left[\frac{(\omega_{\mathrm{c}} + s)(1 + sR_{\mathrm{f}}C_{\mathrm{f}})}{\omega_{\mathrm{c}}R_{\mathrm{f}}} + \frac{s^2(C_{\mathrm{i}} + C_{\mathrm{s}})}{\omega_{\mathrm{c}}}\right]$$

Factor out $1/\omega_{\mathrm{c}}$:

$$I_{\mathrm{pd}} = \frac{V_{\mathrm{out}}}{\omega_{\mathrm{c}}} \left[\frac{(\omega_{\mathrm{c}} + s)(1 + sR_{\mathrm{f}}C_{\mathrm{f}})}{R_{\mathrm{f}}} + s^2(C_{\mathrm{i}} + C_{\mathrm{s}})\right] \tag{9}$$

Step 6: Expand the product.

$$\begin{aligned}(\omega_{\mathrm{c}} + s)(1 + sR_{\mathrm{f}}C_{\mathrm{f}}) &= \omega_{\mathrm{c}} + \omega_{\mathrm{c}}sR_{\mathrm{f}}C_{\mathrm{f}} + s + s^2R_{\mathrm{f}}C_{\mathrm{f}} \\ &= s^2R_{\mathrm{f}}C_{\mathrm{f}} + s(1 + \omega_{\mathrm{c}}R_{\mathrm{f}}C_{\mathrm{f}}) + \omega_{\mathrm{c}}\end{aligned}$$

Step 7: Divide by R_{f} and add the shunt term.

$$\begin{aligned}&\frac{(\omega_{\mathrm{c}} + s)(1 + sR_{\mathrm{f}}C_{\mathrm{f}})}{R_{\mathrm{f}}} + s^2(C_{\mathrm{i}} + C_{\mathrm{s}}) \\ &= s^2C_{\mathrm{f}} + \frac{s(1 + \omega_{\mathrm{c}}R_{\mathrm{f}}C_{\mathrm{f}})}{R_{\mathrm{f}}} + \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}} + s^2(C_{\mathrm{i}} + C_{\mathrm{s}}) \\ &= s^2(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}) + s\,\frac{1 + \omega_{\mathrm{c}}R_{\mathrm{f}}C_{\mathrm{f}}}{R_{\mathrm{f}}} + \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}}\end{aligned}$$

Step 8: Form the transfer function. From (9) and the result of Step 7:

$$Z_{\mathrm{TI}}(s) = \frac{V_{\mathrm{out}}}{I_{\mathrm{pd}}} = \frac{\omega_{\mathrm{c}}}{s^2(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}) + s\,\dfrac{1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}} + \dfrac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}}} \tag{10}$$

(The overall sign is negative for the inverting configuration; we write the magnitude here and note the 180° phase inversion separately.)

Step 9: Identify the standard second-order form. A standard second-order low-pass transfer function has the form:

$$H(s) = \frac{K\,\omega_{\mathrm{n}}^2}{s^2 + 2\zeta\omega_{\mathrm{n}}\,s + \omega_{\mathrm{n}}^2}$$

Dividing numerator and denominator of (10) by $(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})$:

$$Z_{\mathrm{TI}}(s) = \frac{\dfrac{\omega_{\mathrm{c}}}{C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}}}}{s^2 + \dfrac{1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}\,s + \dfrac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}} \tag{11}$$

By comparison with the standard form, we identify:

$$\omega_{\mathrm{n}}^2 = \frac{\omega_{\mathrm{c}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})} \tag{12}$$

$$2\zeta\omega_{\mathrm{n}} = \frac{1 + \omega_{\mathrm{c}} R_{\mathrm{f}} C_{\mathrm{f}}}{R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})} \tag{13}$$

The DC gain ($s = 0$) of (11):

$$Z_{\mathrm{TI}}(0) = \frac{\omega_{\mathrm{c}}/(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})}{\omega_{\mathrm{c}}/[R_{\mathrm{f}}(C_{\mathrm{i}} + C_{\mathrm{f}} + C_{\mathrm{s}})]} = R_{\mathrm{f}}.$$

The numerator constant can therefore be written as $R_{\mathrm{f}}\,\omega_{\mathrm{n}}^2$, giving the final form (restoring the sign for the inverting configuration):

$$Z_{\mathrm{TI}}(s) = \frac{-R_{\mathrm{f}}\,\omega_{\mathrm{n}}^2}{s^2 + 2\zeta\omega_{\mathrm{n}}\,s + \omega_{\mathrm{n}}^2} \tag{14}$$

3 Butterworth-optimum feedback capacitance

Objective. Find the value of C_{f} that makes the closed-loop response maximally flat (Butterworth).

Step 1: Butterworth condition. A second-order Butterworth (maximally flat magnitude) filter has quality factor:

$$Q = \frac{1}{\sqrt{2}}$$

The relationship between Q and the damping ratio ζ is $Q = 1/(2\zeta)$, so:

$$\zeta = \frac{1}{\sqrt{2}} \tag{15}$$

This is the unique damping ratio that produces zero gain peaking in the magnitude response (i.e. $|Z_{\mathrm{TI}}(j\omega)|$ is monotonically decreasing).

Step 2: Translate the condition to circuit parameters. The Butterworth condition (15) implies:

$$(2\zeta\omega_\mathrm{n})^2 = 4\zeta^2\omega_\mathrm{n}^2 = 4 \cdot \frac{1}{2} \cdot \omega_\mathrm{n}^2 = 2\,\omega_\mathrm{n}^2.$$

Substituting (12) and (13):

$$\left(\frac{1+\omega_\mathrm{c}R_\mathrm{f}C_\mathrm{f}}{R_\mathrm{f}(C_\mathrm{i}+C_\mathrm{f}+C_\mathrm{s})}\right)^2 = 2 \cdot \frac{\omega_\mathrm{c}}{R_\mathrm{f}(C_\mathrm{i}+C_\mathrm{f}+C_\mathrm{s})}$$

Multiply both sides by $[R_\mathrm{f}(C_\mathrm{i}+C_\mathrm{f}+C_\mathrm{s})]^2$:

$$(1+\omega_\mathrm{c}R_\mathrm{f}C_\mathrm{f})^2 = 2\,\omega_\mathrm{c}R_\mathrm{f}(C_\mathrm{i}+C_\mathrm{f}+C_\mathrm{s}) \tag{16}$$

Step 3: Solve for C_f. Define a characteristic capacitance:

$$C_\mathrm{c} \equiv \frac{1}{\omega_\mathrm{c}R_\mathrm{f}} = \frac{1}{2\pi f_\mathrm{c}R_\mathrm{f}} \tag{17}$$

This has a physical interpretation: C_c is the capacitance whose impedance at f_c equals R_f.

Rewrite (16) using $\omega_\mathrm{c}R_\mathrm{f} = 1/C_\mathrm{c}$:

$$\left(1+\frac{C_\mathrm{f}}{C_\mathrm{c}}\right)^2 = \frac{2(C_\mathrm{i}+C_\mathrm{f}+C_\mathrm{s})}{C_\mathrm{c}}$$

Expand the left side:

$$1+\frac{2C_\mathrm{f}}{C_\mathrm{c}}+\frac{C_\mathrm{f}^2}{C_\mathrm{c}^2} = \frac{2(C_\mathrm{i}+C_\mathrm{s})}{C_\mathrm{c}}+\frac{2C_\mathrm{f}}{C_\mathrm{c}}$$

The $2C_\mathrm{f}/C_\mathrm{c}$ terms cancel:

$$1+\frac{C_\mathrm{f}^2}{C_\mathrm{c}^2} = \frac{2(C_\mathrm{i}+C_\mathrm{s})}{C_\mathrm{c}}$$

Multiply through by C_c^2:

$$C_\mathrm{f}^2 = 2(C_\mathrm{i}+C_\mathrm{s})\,C_\mathrm{c} - C_\mathrm{c}^2 = C_\mathrm{c}\big(2(C_\mathrm{i}+C_\mathrm{s}) - C_\mathrm{c}\big)$$

Taking the positive root:

$$C_\mathrm{f} = \sqrt{C_\mathrm{c}\big(2(C_\mathrm{i}+C_\mathrm{s}) - C_\mathrm{c}\big)}, \qquad C_\mathrm{c} = \frac{1}{2\pi f_\mathrm{c}R_\mathrm{f}} \tag{18}$$

Validity condition. (18) requires the argument of the square root to be non-negative:

$$2(C_\mathrm{i}+C_\mathrm{s}) - C_\mathrm{c} > 0 \quad\Longleftrightarrow\quad C_\mathrm{i}+C_\mathrm{s} > \frac{C_\mathrm{c}}{2} = \frac{1}{2\omega_\mathrm{c}R_\mathrm{f}}.$$

When this condition is violated, the opamp GBW is too low (or R_f too small) to achieve a Butterworth response—the circuit is fundamentally underdamped regardless of C_f.

Relation to the common approximation. Many references simplify by assuming $\omega_\mathrm{c}R_\mathrm{f}C_\mathrm{f} \gg 1$ (equivalently $C_\mathrm{f} \gg C_\mathrm{c}$). Under this approximation, (16) becomes:

$$(\omega_\mathrm{c}R_\mathrm{f}C_\mathrm{f})^2 \approx 2\,\omega_\mathrm{c}R_\mathrm{f}(C_\mathrm{i}+C_\mathrm{s}) \quad\Longrightarrow\quad C_\mathrm{f} \approx \sqrt{\frac{2(C_\mathrm{i}+C_\mathrm{s})}{\omega_\mathrm{c}R_\mathrm{f}}} = \sqrt{2(C_\mathrm{i}+C_\mathrm{s})\,C_\mathrm{c}}.$$

(18) is the **exact** solution that does not rely on this approximation. The difference is significant when C_c is not negligible compared to $C_\mathrm{i}+C_\mathrm{s}$ (i.e. when R_f is small or f_c is low).

4 Bandwidth

4.1 The $-3\,\mathrm{dB}$ bandwidth of a second-order Butterworth

For a general second-order transfer function with damping ratio ζ, the squared magnitude response is:

$$|H(j\omega)|^2 = \frac{\omega_\mathrm{n}^4}{(\omega_\mathrm{n}^2 - \omega^2)^2 + (2\zeta\omega_\mathrm{n}\omega)^2}$$

For the Butterworth case $\zeta = 1/\sqrt{2}$, substituting and simplifying:

$$|H(j\omega)|^2 = \frac{\omega_\mathrm{n}^4}{(\omega_\mathrm{n}^2 - \omega^2)^2 + 2\omega_\mathrm{n}^2\omega^2}$$

Expand the denominator:

$$= \omega_\mathrm{n}^4 - 2\omega_\mathrm{n}^2\omega^2 + \omega^4 + 2\omega_\mathrm{n}^2\omega^2 = \omega_\mathrm{n}^4 + \omega^4$$

Therefore:

$$|H(j\omega)|^2 = \frac{\omega_\mathrm{n}^4}{\omega_\mathrm{n}^4 + \omega^4} = \frac{1}{1 + (\omega/\omega_\mathrm{n})^4} \tag{19}$$

This is the defining form of a second-order Butterworth response.

At $\omega = \omega_\mathrm{n}$:

$$|H(j\omega_\mathrm{n})|^2 = \frac{1}{2} \quad \Longrightarrow \quad |H(j\omega_\mathrm{n})| = \frac{1}{\sqrt{2}}\ (-3\,\mathrm{dB}).$$

Therefore the $-3\,\mathrm{dB}$ bandwidth is exactly the natural frequency:

$$\omega_{-3\,\mathrm{dB}} = \omega_\mathrm{n}.$$

In terms of ordinary frequency:

$$f_{-3\,\mathrm{dB}} = \frac{\omega_\mathrm{n}}{2\pi} \tag{20}$$

4.2 Express in terms of circuit parameters

From (12):

$$\omega_\mathrm{n} = \sqrt{\frac{\omega_\mathrm{c}}{R_\mathrm{f}(C_\mathrm{i} + C_\mathrm{f} + C_\mathrm{s})}} = \sqrt{\frac{2\pi f_\mathrm{c}}{R_\mathrm{f}(C_\mathrm{i} + C_\mathrm{f} + C_\mathrm{s})}}$$

Therefore:

$$f_{-3\,\mathrm{dB}} = \frac{\omega_\mathrm{n}}{2\pi} = \sqrt{\frac{f_\mathrm{c}}{2\pi R_\mathrm{f}(C_\mathrm{i} + C_\mathrm{f} + C_\mathrm{s})}} \tag{21}$$

5 Noise equivalent bandwidth

5.1 Definition

The noise equivalent bandwidth (NEB) of a filter $H(f)$ is the bandwidth of an ideal rectangular (brick-wall) filter with the same peak gain that passes the same total noise power:

$$\mathrm{NEB} = \frac{1}{|H_\mathrm{max}|^2} \int_0^\infty |H(f)|^2\,\mathrm{d}f$$

For a Butterworth filter, $|H_\mathrm{max}| = |H(0)| = 1$, so:

$$\mathrm{NEB} = \int_0^\infty |H(f)|^2\,\mathrm{d}f \tag{22}$$

5.2 General n-th order Butterworth NEB

For an n-th order Butterworth filter, the squared magnitude response is:

$$|H(f)|^2 = \frac{1}{1 + (f/f_{-3\,\mathrm{dB}})^{2n}}$$

Substituting into (22) with $u = f/f_{-3\,\mathrm{dB}}$:

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \int_0^\infty \frac{\mathrm{d}u}{1 + u^{2n}} \tag{23}$$

The integral $I_n = \displaystyle\int_0^\infty \frac{\mathrm{d}u}{1 + u^{2n}}$ is a standard result that can be evaluated using the substitution $t = u^{2n}$ and recognizing the beta function.

Setting $t = u^{2n}$, so $u = t^{1/(2n)}$, $\mathrm{d}u = \frac{1}{2n}\, t^{1/(2n)-1}\, \mathrm{d}t$:

$$I_n = \int_0^\infty \frac{1}{1+t} \cdot \frac{1}{2n}\, t^{1/(2n)-1}\, \mathrm{d}t = \frac{1}{2n}\, B\!\left(\frac{1}{2n},\, 1 - \frac{1}{2n}\right)$$

where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a + b)$ is the beta function.

Using the reflection formula $\Gamma(z)\,\Gamma(1 - z) = \pi/\sin(\pi z)$ with $z = 1/(2n)$:

$$I_n = \frac{1}{2n} \cdot \frac{\pi}{\sin\!\left(\frac{\pi}{2n}\right)} \tag{24}$$

Therefore:

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2n\,\sin\!\left(\frac{\pi}{2n}\right)} \tag{25}$$

5.3 Specialization to $n = 2$

For the second-order Butterworth ($n = 2$):

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{4 \cdot \frac{\sqrt{2}}{2}} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2\sqrt{2}} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi\sqrt{2}}{4}$$

$$\mathrm{NEB} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi\sqrt{2}}{4} \approx 1.111\, f_{-3\,\mathrm{dB}} \tag{26}$$

5.4 Comparison with the single-pole approximation

For a single-pole (first-order) system ($n = 1$):

$$\mathrm{NEB}_{n=1} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2\sin(\pi/2)} = f_{-3\,\mathrm{dB}} \cdot \frac{\pi}{2} \approx 1.571\, f_{-3\,\mathrm{dB}}.$$

Many TIA references use this single-pole approximation even when the circuit has a second-order response. The second-order Butterworth NEB is $1.111/1.571 = 0.707$ times the single-pole value—a 29% difference that directly affects computed noise levels.

6 Noise analysis

The TIA output noise is the root-sum-square (RSS) of several independent noise sources. Each source has a spectral density that is shaped by the closed-loop gain before appearing at the output. The noise sources fall into two categories: (a) voltage noise of the opamp, which is amplified by the frequency-dependent noise gain, and (b) current/thermal noise sources whose spectra are flat and are shaped only by the closed-loop transimpedance $|Z_{\mathrm{TI}}(f)|^2$.

6.1 Noise gain

The noise gain $G_\mathrm{n}(f)$ is the closed-loop gain from the opamp's input voltage noise source e_ni to the output. For a TIA with feedback impedance $Z_\mathrm{f} = R_\mathrm{f} \| (1/sC_\mathrm{f})$ and total input-to-ground capacitance $(C_\mathrm{i} + C_\mathrm{s})$, the noise gain is:

$$G_\mathrm{n}(f) = 1 + \frac{Z_\mathrm{in}}{Z_\mathrm{f}} = 1 + \frac{C_\mathrm{i} + C_\mathrm{s}}{C_\mathrm{f} + C_\mathrm{s}} \cdot \frac{1 + sR_\mathrm{f}(C_\mathrm{f} + C_\mathrm{s})}{1 + sR_\mathrm{f}(C_\mathrm{i} + C_\mathrm{f} + C_\mathrm{s})}$$

This gain exhibits four distinct frequency regions that are exploited for piecewise integration:

1. **Low-frequency flat region** ($f < f_\mathrm{zf}$): $G_\mathrm{n} \approx 1$
2. **Rising region** ($f_\mathrm{zf} < f < f_\mathrm{pf}$): $G_\mathrm{n} \approx f/f_\mathrm{zf}$
3. **Plateau region** ($f_\mathrm{pf} < f < f_\mathrm{i}$): $G_\mathrm{n} \approx 1 + C_\mathrm{i}/(C_\mathrm{f} + C_\mathrm{s})$
4. **Roll-off region** ($f > f_\mathrm{i}$): G_n decreases as the loop gain falls below unity

where the characteristic frequencies are:

$$f_\mathrm{zf} = \frac{1}{2\pi R_\mathrm{f}(C_\mathrm{i} + C_\mathrm{f} + C_\mathrm{s})} \tag{27}$$

$$f_\mathrm{pf} = \frac{1}{2\pi R_\mathrm{f}(C_\mathrm{f} + C_\mathrm{s})} \tag{28}$$

$$f_\mathrm{i} = f_\mathrm{c} \cdot \frac{C_\mathrm{f} + C_\mathrm{s}}{C_\mathrm{i} + C_\mathrm{f} + C_\mathrm{s}} \tag{29}$$

Here f_zf is the noise-gain zero frequency, f_pf is the noise-gain pole frequency, and f_i is the unity loop-gain (intersection) frequency.

6.2 Opamp voltage noise (piecewise integration)

The opamp input voltage noise spectral density has a $1/f$ component below the corner frequency f_f and a white (flat) floor e_nif above it:

$$e_\mathrm{ni}(f) = e_\mathrm{nif}\sqrt{1 + f_\mathrm{f}/f} \tag{30}$$

The output-referred voltage noise RMS is obtained by integrating the product of $e_\mathrm{ni}^2(f)$ and $G_\mathrm{n}^2(f)$ over frequency, divided into five segments. Defining a low-frequency cutoff f_1 (set to 0.01 Hz in the implementation):

Region 1—$1/f$ noise (f_1 to f_f), noise gain ≈ 1:

$$E_\mathrm{noe,1}^2 = e_\mathrm{nif}^2 \cdot f_\mathrm{f} \ln\left(\frac{f_\mathrm{f}}{f_1}\right) \tag{31}$$

Region 2—white noise, flat gain (f_f to f_zf), noise gain ≈ 1:

$$E_\mathrm{noe,2}^2 = e_\mathrm{nif}^2(f_\mathrm{zf} - f_\mathrm{f}) \tag{32}$$

Region 3—white noise, rising gain (f_zf to f_pf), noise gain $\approx f/f_\mathrm{zf}$:

$$E_\mathrm{noe,3}^2 = \frac{e_\mathrm{nif}^2}{f_\mathrm{zf}^2} \cdot \frac{f_\mathrm{pf}^3 - f_\mathrm{zf}^3}{3} \tag{33}$$

Region 4—white noise, plateau (f_{pf} to f_{i}), noise gain $\approx 1 + C_{\mathrm{i}}/(C_{\mathrm{f}} + C_{\mathrm{s}})$:

$$E^2_{\mathrm{noe},4} = e^2_{\mathrm{nif}} \left(1 + \frac{C_{\mathrm{i}}}{C_{\mathrm{f}} + C_{\mathrm{s}}}\right)^2 (f_{\mathrm{i}} - f_{\mathrm{pf}}) \tag{34}$$

Region 5—above unity loop gain ($f > f_{\mathrm{i}}$), gain $\approx f_{\mathrm{c}}/f$:

$$E^2_{\mathrm{noe},5} = e^2_{\mathrm{nif}} \cdot \frac{f_{\mathrm{c}}^2}{f_{\mathrm{i}}} \tag{35}$$

Note: (35) arises from $\int_{f_{\mathrm{i}}}^{\infty} (f_{\mathrm{c}}/f)^2 \, \mathrm{d}f = f_{\mathrm{c}}^2/f_{\mathrm{i}}$.

The total output-referred opamp voltage noise is:

$$E_{\mathrm{noe}} = \sqrt{E^2_{\mathrm{noe},1} + E^2_{\mathrm{noe},2} + E^2_{\mathrm{noe},3} + E^2_{\mathrm{noe},4} + E^2_{\mathrm{noe},5}} \tag{36}$$

6.3 Feedback resistor thermal noise

The Johnson noise of R_{f} produces a current noise density $\sqrt{4k_{\mathrm{B}}T/R_{\mathrm{f}}}$, which flows through R_{f} to produce output voltage noise. Since this noise source sees the same transimpedance transfer function as the signal, the NEB ((26)) is the appropriate bandwidth:

$$E_{\mathrm{noR}} = \sqrt{4k_{\mathrm{B}}TR_{\mathrm{f}} \cdot \mathrm{NEB}} \tag{37}$$

6.4 Shot noise

Shot noise from DC currents (bias current I_{b}, dark current I_{d}, and signal photocurrent I_{p}) produces white current noise with spectral density $\sqrt{2qI}$. Each is converted to output voltage noise by the transimpedance R_{f} over the NEB:

$$E_{\mathrm{noi,bias}} = R_{\mathrm{f}} \sqrt{2q\, I_{\mathrm{b}} \cdot \mathrm{NEB}} \tag{38}$$

$$E_{\mathrm{noi,dark}} = R_{\mathrm{f}} \sqrt{2q\, I_{\mathrm{d}} \cdot \mathrm{NEB}} \tag{39}$$

$$E_{\mathrm{noi,sig}} = R_{\mathrm{f}} \sqrt{2q\, I_{\mathrm{p}} \cdot \mathrm{NEB}} \tag{40}$$

6.5 Opamp current noise

The opamp input current noise floor i_{nif} (white spectral density) also flows through R_{f}:

$$E_{\mathrm{noi,nif}} = R_{\mathrm{f}}\, i_{\mathrm{nif}} \sqrt{\mathrm{NEB}} \tag{41}$$

6.6 Total noise and signal-to-noise ratio

The total background noise (all sources except signal shot noise) is:

$$E_{\mathrm{no,bg}} = \sqrt{E^2_{\mathrm{noe}} + E^2_{\mathrm{noR}} + E^2_{\mathrm{noi,bias}} + E^2_{\mathrm{noi,dark}} + E^2_{\mathrm{noi,nif}}} \tag{42}$$

The signal voltage is $V_{\mathrm{sig}} = R_{\mathrm{f}} \cdot I_{\mathrm{p}} = R_{\mathrm{f}} \cdot r_{\phi} \cdot P_{\mathrm{opt}}$, where r_{ϕ} is the photodiode responsivity and P_{opt} is the incident optical power. The signal-to-noise ratio is:

$$\mathrm{SNR} = \frac{V_{\mathrm{sig}}}{\sqrt{E^2_{\mathrm{no,bg}} + E^2_{\mathrm{noi,sig}}}} \tag{43}$$

The ratio $E_{\mathrm{no,bg}}/E_{\mathrm{noi,sig}}$ indicates whether the system is shot-noise limited (ratio < 1) or background-noise limited (ratio > 1).

7 Implementation mapping

Theory	Implementation	File
Opamp model, (2)	`Opamp` dataclass (`f_c` field)	`tia_design.py`
Input capacitance C_i	`C_i = C_d + opamp.C_icm + opamp.C_id`	`tia_design.py`
Butterworth C_f, (18)	`C_f = (C_c * (2*(C_i+C_s) - C_c))**0.5`	`tia_design.py`
Bandwidth, (21)	`BW_t = (f_c / (2*pi*R_f*C_tot))**0.5`	`tia_design.py`
NEB, (26)	`NEB = (pi*sqrt(2)/4) * BW_t`	`tia_design.py`
Noise gain frequencies, (27)–(29)	`f_zf`, `f_pf`, `f_i`	`tia_design.py`
Piecewise voltage noise, (31)–(36)	`E_noe1` ... `E_noe5`, `E_noe`	`tia_design.py`
Thermal noise, (37)	`E_noR`	`tia_design.py`
Shot noise, (38)–(40)	`E_noi_bias`, `E_noi_dark`, `E_noi_sig`	`tia_design.py`
Current noise, (41)	`E_noi_nif`	`tia_design.py`
SNR, (43)	`snr`	`tia_design.py`
Photodiode $C_\mathrm{d}(V_\mathrm{b})$ model	`Photodiode.C_d(Vb)`	`tia_design.py`
Multi-opamp comparison	`compare_opamps(...)`	`tia_design.py`

8 References

1. J. Graeme, *Photodiode Amplifiers: Op Amp Solutions*. McGraw-Hill, 1996.
2. P. C. D. Hobbs, *Building Electro-Optical Systems: Making It All Work*, 2nd ed. Wiley, 2009.

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